COMPUTER GRAPHICS PROJECT

Bloxorz 3D

Three.js and WebGL puzzle game

PROJECT OVERVIEW

Playable puzzle, complete flow

The project contains eight verified levels, a rolling cuboid, fragile tiles, switches, bridges, teleport split mode, saved progress, a custom editor, and a hint system based on shortest-path search.

CORE MODEL

The puzzle state is discrete

The state is x, z, and orientation. Standing occupies one tile, lying_x occupies two tiles on the x axis, and lying_z occupies two tiles on the z axis. Those occupied cells decide legal moves, falling, switches, bridges, and winning.

Smooth motion follows exact rules

Each input changes an exact grid state first. The mesh then interpolates its position, rotates with quaternion motion, and snaps back to the target coordinate so the animation stays smooth without damaging the puzzle logic.

LEVEL MECHANICS

Rules on the board

Normal tiles support the block. Fragile tiles break under a standing block. Soft switches react to any occupied cell, while heavy switches require the full upright block. Bridges change the level graph, teleport splits the block into two cubes, and the goal accepts only a standing full block.

Course concepts in the scene

The implementation moves from geometry to model transforms, then to the camera view transform, perspective projection, rasterization, fragment shading, and finally the browser canvas frame buffer.

RENDERING

Scene and visual feedback

The scene uses PerspectiveCamera, MeshPhysicalMaterial, directional and ambient lighting, shadows, fog, SMAA, bloom, vignette, procedural sky, particles, procedural audio, and raycasting for editor tile placement.

One game state, multiple controls

Keyboard controls, touch controls, the third-person camera, the HUD, hints, and localStorage progress all read from the same game state. This keeps desktop, mobile, and saved progress consistent.

SOLVER

BFS over puzzle states

Breadth-first search explores moves level by level, so the first solution found is shortest. The state key includes position, orientation, bridge state, and split-block state for verification.

VERIFIED LEVELS

Par values checked by script

First Steps is 8, Over the Edge is 7, Fragile Ground is 6, Open Sesame is 6, Heavy Duty is 12, Split Decision is 8, Twin Bridges is 13, and The Gauntlet is 15.

The command npm run check:levels verifies that every official level solves with its listed par value.

CODE STRUCTURE

Main files

main.js controls flow, grid.js builds the board, controls.js handles movement, levels.js stores level data, solver.js performs BFS, editor.js creates custom levels, and scene/camera/lighting files define the visual world.

LIVE DEMO

Five-minute flow

The demo starts with level selection, then shows Level 1 movement, a fragile tile, a switch and bridge, teleport split and merge, one BFS hint, and the editor if time remains.

SUMMARY

What the project demonstrates

The project demonstrates real-time 3D rendering, model-view-projection concepts, lighting and shading, event-driven interaction, camera control, state-space search, persistence, and a complete playable workflow.